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A novel model predictive dynamic adaptive impedance control for robotic force tracking on complex surfaces

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ABSTRACT

The time-varying nature and complexity of environmental stiffness make it difficult to achieve accurate and stable tracking of normal contact force on complex surfaces in robotic belt grinding. The more complex the contact environment, the higher the requirements for robotic perception and self-regulation capabilities in relation to the environment. To address this issue, a novel model predictive dynamic adaptive impedance control algorithm (MPRPO-DAIC) is proposed to improve the accuracy of robotic force tracking in complex contact environments. This control strategy consists of a dynamic adaptive impedance control algorithm integrated with a model predictive control framework. The dynamic adaptive impedance control enhances the robustness of adaptive impedance controller (AIC) by introducing a dynamic observer term. Additionally, the rolling parameter optimization algorithm enables real-time optimization of key parameters within the dynamic observer. Subsequently, through the combination of dynamic adaptive impedance control and model predictive control, the robotic velocity compensation value is calculated by solving a quadratic programming problem. Finally, the velocity compensation value is integrated and sent to the robotic end-effector, enabling stable tracking of contact forces in complex surfaces. The simulation and experimental results of force tracking on complex surfaces validate the superiority of the proposed control strategy. The simulation results show that MPRPO-DAIC exhibits optimal force control performance in complex tracking environments. In the force tracking experiments, MPRPO-DAIC improved the force control accuracy by 37.25% and 44.90% compared to AIC on surfaces with large deformation and high-frequency variation, respectively.

1. Introduction

Robotic belt grinding, with its advantages of high flexibility, easy integration, and low cost, has gradually become an important method for processing complex surfaces (Ke et al., 2023; Zhou & Tian, 2024). As a typical contact operation, robotic grinding requires contact between the robot and the workpiece, making position control alone insufficient to meet the task requirements (Xu, Zhu, Zhang, Yan, & Ding, 2019a). To address this issue, robotic compliance control (Han et al., 2023; Zhu, Cai, et al., 2023) has been proposed. However, the robotic low stiffness, as well as the time-varying nature and complexity of environmental stiffness during complex surface tracking (Wang et al., 2023a), can affect the accuracy and stability of contact forces during the complex surface tracking process (Mu, Wang, Zou, Wang, Lv, & Wang, 2023).

Hybrid force/position control is a crucial framework for contact force control, which divides the task space into a position tracking subspace and a force tracking subspace. To achieve more flexible control of the robotic end-effector, many researchers have developed force control algorithms within this framework to improve the force control accuracy. Xu, Chen, Zhu, Yan, and Ding (2021a) proposed a hybrid force control strategy using PD control for position and PI control for force, effectively reducing grinding marks after grinding. Chen et al. (2018) designed a two-degree-of-freedom end-effector to achieve force/position control in robotic grinding, employing a PI control structure for both force and position control components. Yun et al. (2022) designed the fuzzy-PID-based position controller, achieving effective force precision and trajectory tracking, with simulations showing enhanced compliance and control performance. Zhang, Li, Zhao, and Zhao (2021) proposed a hybrid anti-disturbance control strategy that combines fuzzy PID and velocity gain loops, enhancing the stability of robot grinding systems. Wang, Zou, Su, Luo, Li, and Huang (2021) designed a hybrid force/position controller based on adaptive fuzzy control to improve both force precision and position tracking for robotic manipulators in uncertain environments. This method does not consider the dynamic coupling between the robot and the contact environment, so it achieves good tracking accuracy under favorable force feedback conditions. However, its primary limitation lies in the

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inability to simultaneously control position and force in the same constrained direction.

To address this, many researchers have combined force/ position hybrid control strategies with impedance control strategies to develop hybrid impedance control strategies. Duan, Gan, Chen, and Dai (2018) proposed an adaptive impedance control method that not only tracks dynamic desired forces but also compensates for environmental uncertainties, significantly enhancing the force-tracking performance of the impedance control method. Sheng and Zhang (2018) introduced a fuzzy adaptive controller that optimizes key parameters in the adaptive impedance controller based on fuzzy rules, effectively reducing contact force errors caused by variations in environmental stiffness. Labrecque and Gosselin (2018) investigated the adaptive and optimized impact of variable admittance parameters on system dynamic response, proposing three interdependent adaptive control laws to achieve compliant control. Kim, Kim, and Ryu (2014) proposed a novel adaptive energy-bounding approach (EBA) that ensures robust and stable contact under significant environmental uncertainties while maintaining the desired steady-state contact force. Arnold, Hanzlick, and Lee (2019) introduced a variable damping control strategy to determine damping values at a robot ankle joint, ranging from negative to positive, significantly enhancing the agility and performance of human–robot systems. Wu et al. (2018) designed an adaptive impedance controller based on a neural network online compensation algorithm to enhance the regional force control accuracy during the blade grinding process. Hamedani, Sadeghian, Zekri, Sheikholeslam, and Keshmiri (2021) proposed an Intelligent Impedance Control approach based on a Wavelet Neural Network (IIC-WNN) to adaptively adjust impedance parameters in unknown environments, enhancing force tracking performance. From the above studies, it can be seen that research on adaptive methods based on traditional impedance control plays an important role in addressing environmental changes, enhancing robotic compliance, and improving force tracking accuracy.

In complex surface tracking tasks, although adaptive strategies provide a certain level of compensation, their effectiveness is limited due to the dynamic variation and complexity of environmental stiffness. To address this, many researchers adopt Moving Horizon Estimation (rolling estimation strategy) to optimize model parameters or predict currently unmeasurable states in real-time. Zhang, Zhao, Chen, Tao, and Ding (2024) proposed an observer-based variable impedance control (MHE-VIC) approach, leveraging moving horizon estimation to dynamically adjust impedance based on real-time observations, enhancing control precision and stability in thin-walled workpiece machining. Akdoğan, Aktan, Koru, Arslan, Atlıhan, and Kuran (2018) proposed a moving horizon estimation (MHE)-based approach for sensor reduction in variable stiffness actuated (VSA) robots. This framework optimizes control by estimating unmeasured states in real-time, enhancing stability and reliability in human–robot interaction tasks. Schiller and Müller (2023) proposed a suboptimal MHE method, enhancing stability and robustness in nonlinear systems with computational efficiency for real-time applications. Liu, Zhang, Chen, and Yu (2017) proposed a multirate MHE method that fills slower sensor data gaps with predictions, enhancing continuity and improving accuracy and stability in real-time estimation. Liu, Yu, and Zhang (2014) proposed a Moving Horizon Estimation (MHE) method that modeled packet loss as a Bernoulli process and employed a one-step MHE algorithm to enhance the accuracy and robustness of signal-to-interference-plus-noise ratio (SINR) estimation in wireless networks. It can be observed that an appropriate rolling optimization strategy can improve the stability, accuracy, and reliability of the system to a certain extent.

Model predictive control, as an advanced control method, can predict the evolution of system variables over a certain time horizon by measuring the current state of the system, and determine the optimal input for a specific performance index (dos Santos & Siqueira, 2019; Nurbayeva & Rubagotti, 2025). Therefore, many researchers combine controllers with MPC to enhance their performance. Bednarczyk, Omran, and Bayle (2020) presented a model predictive control (MPC) method that facilitates compliant behavior in robot-environment interactions while adhering to constraints. Validation through experiments highlights its effectiveness in managing speed and acceleration limits, enhancing operator safety. Gold, Volz, and Graichen (2023) introduced an MPC-based interaction control framework (MPIC) for robotic arms, acting as a trajectory planner between task and robot controllers. By predicting external forces, MPIC enables flexible motion, force, and compliance control across various manipulation tasks. Jin, Zhang, Liu, and Xu (2023) proposed a model predictive tracking control framework for end-effector pose regulation, aiming to minimize tracking errors as well as the norms of joint velocity and acceleration. Kleff, Dantec, Saurel, Mansard, and Righetti (2022) proposed an innovative MPC method using force feedback to optimize robot performance in contact tasks, integrating joint torque dynamics for real-time control adjustments, enhancing task accuracy and stability. Minniti, Grandia, Fah, Farshidian, and Hutter (2021) proposed an innovative MPC approach for mobile manipulators, combining adaptive control and system identification to enhance robot-environment interaction in unmodeled scenarios. Wijayarathne, Zhou, Zhao, and Hammond (2023) present a real-time MPC framework integrating soft-contact mechanics, optimizing robot stability and precision in flexible, deformable environments like medical applications. Zhu, Liu, and Yu (2023) proposed an LSTM-based learning MPC framework that integrates nominal and learning models to enhance robot robustness and accuracy in trajectory tracking under model uncertainties. Beck, Vu, Hartl-Nesic, and Kugi (2024) proposed a novel waypoint MPC (wMPC) approach, enabling dynamic trajectory optimization by incorporating dynamically changing waypoints, enhancing online replanning efficiency and accuracy in multi-waypoint tasks. The above studies have optimized or enhanced the capability of robotic trajectory tracking or force tracking to a certain extent by integrating MPC. The key focus of using the MPC framework lies in establishing a reasonable and accurate predictive model. This is particularly important for the design of force controllers within the MPC framework in complex contact environments.

Considering the above issues, this paper proposes a novel model predictive dynamic adaptive impedance controller. The main contributions of this paper are as follows:

(1) The dynamic observer-based adaptive impedance controller (DO-AIC) is established. A dynamic observer term is added to the adaptive impedance control, enabling real-time adjustment of key parameters in the adaptive impedance controller (AIC) based on contact force errors and their rate of change, thereby enhancing the robustness of AIC in complex environments. Subsequently, a stability proof is provided to determine the stability boundaries of the DO-AIC.

(2) The key parameters in the dynamic observer are optimized using a rolling optimization algorithm. The gradient descent method is used to determine the sensitivity of the force error to the key parameters. Using this gradient information, quadratic optimization is performed at each time step. By solving a quadratic programming problem, the optimal parameter values for each control cycle are obtained, achieving real-time parameter optimization.

(3) The MPRPO-DAIC algorithm is developed by integrating dynamic adaptive impedance control with model predictive control. The cost function in the model is constructed with the objective of minimizing the contact force error. Subsequently, the velocity compensation value is integrated to compensate for the robotic end-effector, enabling stable tracking of contact forces on complex surfaces.

This paper is structured as follows. Section 2 presents the force tracking framework in robotic grinding. Section 3 introduces a novel model predictive dynamic adaptive impedance control (MPRPO-DAIC) algorithm for grinding force tracking tasks on complex surfaces. In Section 4, the effectiveness of the proposed controller is validated through force tracking simulation experiments. Finally, Section 5 presents force tracking experiments on complex surfaces to evaluate the effectiveness and robustness of the proposed method.

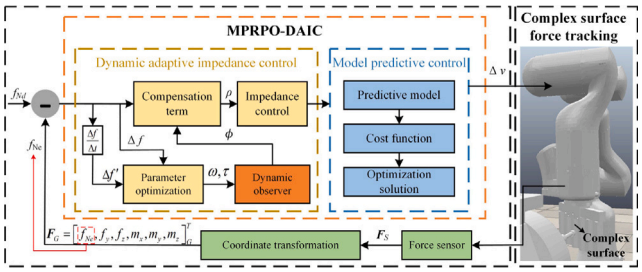


Fig. 1. Overview of robotic force tracking process framework .

2. Robotic force tracking framework

In the robotic grinding process on complex surfaces, the quality of normal contact force control directly affects the material removal accuracy (Li, Yang, Zhao, & Ding, 2022; Song, Xiao, Chen, Liu, & Huang, 2023), surface quality (Xu, Chen, Zhu, Yan, & Ding, 2021b; Zhang et al., 2020), and overall machining precision of the workpiece (Mu, Zou, Wang, Li, Yan, & Wang, 2025; Wang et al., 2023b; Xu, Zhu, Zhang, Yan, & Ding, 2019b). Therefore, this paper primarily focuses on the control of normal contact force during the robotic grinding process. Fig. 1 illustrates the framework of the robotic normal contact force tracking process. The contact force errors generated during the interaction between the robot and the complex surface are used as inputs to the MPRPO-DAIC, which computes the displacement compensation values for the robotic end-effector. The MPRPO-DAIC is specifically implemented by integrating dynamic adaptive impedance control with model predictive control. Dynamic adaptive impedance control improves the robustness of the adaptive impedance controller (AIC) by adding a dynamic observer term. Furthermore, the key parameters within the dynamic observer are optimized using a rolling parameter optimization method. Based on dynamic adaptive impedance control, model predictive control is applied, and the velocity compensation value is obtained by solving a quadratic programming problem (Gold, Völz, & Graichen, 2020). The velocity compensation value is then converted into a displacement compensation value for the robotic end-effector through integration.

As shown in Fig. 2, the coordinate systems for the robot base, flange, sensor, workpiece, grinding head, and cutter-contact (CC) point are denoted as $\{B\},\{F\},\{S\},\{W\},\{G\}$, and $\{C\}$, respectively. The process chain of robotic contact force tracking on complex surfaces can be expressed as:

$$ {}^{F}_{B}\boldsymbol{T} = {}^{G}_{B}\boldsymbol{T}\left({}^{G}_{C}\boldsymbol{T}\right)^{-1}\left({}^{C}_{W}\boldsymbol{T}\right)^{-1}\left({}^{W}_{S}\boldsymbol{T}\,{}^{S}_{F}\boldsymbol{T}\right)^{-1} \tag{1} $$

The transformation matrix ${}^{F}_{B}\boldsymbol{T}$ determines the robotic spatial position. The transformation matrix ${}^{G}_{B}\boldsymbol{T}$ is used to determine the position of the coordinate system {G} relative to the {B}, which can be obtained through probe calibration (Mu et al., 2023). The ${}^{G}_{C}\boldsymbol{T}$ is a homogeneous transformation matrix, where the rotation component aligns the coordinate system {C} with {G}. The transformation matrix ${}^{W}_{F}\boldsymbol{T}$ is used to determine the position of the coordinate system {W} relative to the {F}. Since a sensor is installed between the workpiece and the flange, ${}^{W}_{F}\boldsymbol{T}$ can be expressed as ${}^{W}_{S}\boldsymbol{T}\,{}^{S}_{F}\boldsymbol{T}$.

The transformation matrix ${}^{C}_{W}\boldsymbol{T}$ represents the coordinate of the CC point $\boldsymbol{c}_i$ on the curved workpiece relative to the coordinate system {W} and can be expressed as:

$$ {}^{C}_{W}\boldsymbol{T} = \begin{bmatrix} u_{xi} & v_{xi} & n_{xi} & x_i \\ u_{yi} & v_{yi} & n_{yi} & y_i \\ u_{zi} & v_{zi} & n_{zi} & z_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2} $$

Here, (x_i, y_i, z_i) represents the spatial position coordinates of the CC point on the complex surface, $\boldsymbol{u}_i = \left[u_{xi}, u_{yi}, u_{zi}\right]^T$ and $\boldsymbol{v}_i = \left[v_{xi}, v_{yi}, v_{zi}\right]^T$ represent the u-direction and v-direction of the CC point on the complex surface, respectively, where $\boldsymbol{v}_i = \boldsymbol{u}_i \times \boldsymbol{n}$.

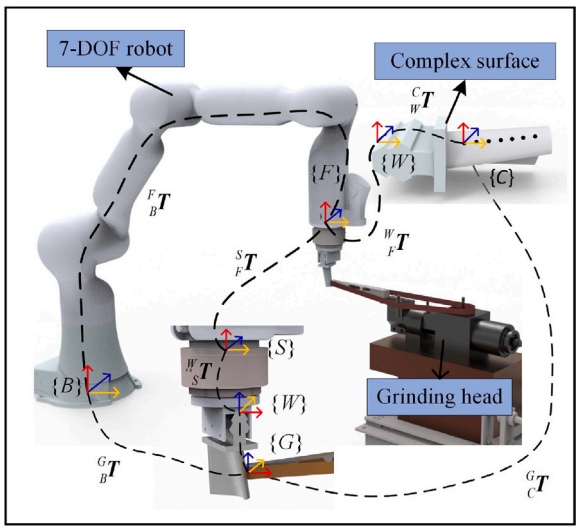


Fig. 2. Complex surface tracking process chain.

The force signals collected by the force/torque sensor are represented in the coordinate system {S} as $\boldsymbol{F}_S = \left[f_x, f_y, f_z, m_x, m_y, m_z\right]^T_S$. The force signal $\boldsymbol{F}_G$ in the coordinate system {G} is computed using the adjoint transformation matrix as follows:

$$ \begin{cases} \boldsymbol{F}_G = Ad_F\left({}^{G}_{S}\boldsymbol{T}\right)\boldsymbol{F}_S \\ Ad_F\left({}^{G}_{S}\boldsymbol{T}\right) = \begin{bmatrix} {}^{G}_{S}\boldsymbol{R} & 0 \\ \left[{}^{G}_{S}\boldsymbol{g}\right]{}^{G}_{S}\boldsymbol{R} & {}^{G}_{S}\boldsymbol{R} \end{bmatrix} \end{cases} \tag{3} $$

The adjoint matrix $Ad_F\left({}^{G}_{S}\boldsymbol{T}\right) \in \mathbb{R}^{6\times6}$, where ${}^{G}_{S}\boldsymbol{R} \in \mathbb{R}^{3\times3}$ is the rotation matrix, and ${}^{G}_{S}\boldsymbol{g} \in \mathbb{R}^{3\times1}$ represents the displacement vector of the coordinate system {S} relative to the coordinate system {G}. The term $\left[{}^{G}_{S}\boldsymbol{g}\right]$ is the skew-symmetric matrix of ${}^{G}_{S}\boldsymbol{g}$. $\boldsymbol{F}_G$ represents the actual force signals in the coordinate system {G} and are expressed as $\boldsymbol{F}_G = \left[f_x, f_y, f_{Ne}, m_x, m_y, m_z\right]^T_G$. Here, the z-axis of the coordinate system {G} corresponds to the direction of the normal contact force, so f_{Ne} is the actual contact force to be tracked.

3. MPRPO-DAIC algorithm

The MPRPO-DAIC algorithm consists of the dynamic adaptive impedance control (Section 3.2) integrated with model predictive control, and the overall control flow of the MPRPO-DAIC algorithm is described in the model predictive dynamic adaptive impedance control algorithm (Section 3.3).

3.1. Impedance model between the robot and the environment

Robotic grinding is a process transitioning from the free motion phase to the grinding contact phase. Phase I is the free motion phase before the robot end-effector contacts the environment, during which the robot has not made contact with the workpiece. In Phase II, the robot end-effector collides with the environment, resulting in a brief contact oscillation. By Phase III, stable contact is established between the robot end-effector and the environment (Duan et al., 2018). As illustrated in Fig. 3, z_c denotes the position command provided to the robot, and z_d indicates the desired position of the robot.

The robotic contact force control mainly focuses on the grinding contact phases (Phase II and Phase III). Here, a spring–mass–damper assumption is made along the normal grinding force direction, with the environment modeled as a spring of stiffness k_e. The motion differential

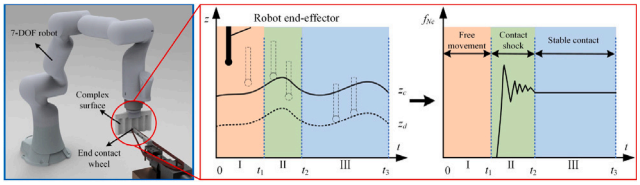


Fig. 3. Interaction process between the robot and environment.

equation of the entire grinding system along the normal grinding force direction can be expressed as:

$$\begin{cases} m_r\ddot{z}(t) + b_r\dot{z}(t) + k_r z(t) = 0, & z(t) > l \\ m_r\ddot{z}(t) + b_r\dot{z}(t) + k_r z(t) = k_e(l - z(t)), & z(t) \le l \\ z\left(t_1\right) = l \end{cases} \tag{4}$$

Here, the robotic reference mass m_r, damping b_r, stiffness k_r, and the environmental stiffness k_e are scalar parameters defined along the normal grinding force direction. At time t_1, the robot reaches the initial contact position l along the normal grinding force direction. The robot end-effector pose is defined as $\boldsymbol{P} = [x, y, z, w, p, r]^T$, where the Euler angles w, p, r represent the orientation of the robotic end-effector in the Cartesian coordinate system. In the contact model, $z(t) \in \boldsymbol{P}(t)$ denotes the translational displacement along the normal grinding force direction, and the corresponding motion equation is given by:

$$m_r\ddot{z}(t) + b_r\dot{z}(t) + \left(k_r + k_e\right) z(t) = k_e l \tag{5}$$

The position tracking error of the robot along the normal grinding force direction at time t can be expressed as $e(t) = z(t) - z_d(t)$, where $z_d(t)$ denotes the desired position at time t. In this case, the motion differential equation can be rewritten in the form of the force tracking impedance equation, as illustrated in Fig. 4, which is specifically described as:

$$e = f_{Ne} - f_{Nd} = m_r\left(\ddot{z} - \ddot{z}_d\right) + b_r\left(\dot{z} - \dot{z}_d\right) + k_r\left(z - z_d\right) \tag{6}$$

The contact force between the robot end-effector and the external environment is represented as $f_{Ne} = k_e\left(z_c - z_d\right)$, and the tracking error of the robot control system is ignored, then $z = z_c$. Based on Eq. (6), the transfer function from the displacement error $z(t) - z_d(t)$ to the force tracking error $\Delta f(t)$ can be expressed in the Laplace domain as: $k(s) = \Delta f(s)/\left(z(s) - z_d(s)\right) = 1/\left(m_r s^2 + b_r s + k_r\right)$. Accordingly, the following equation is derived:

$$\begin{aligned} &\Delta f(s)\left(m_r s^2 + b_r s + k_r + k_e\right) \\ &= \left(m_r s^2 + b_r s + k_r\right)\left[k_e\left(z - z_d\right) - f_{Nd}\right] \end{aligned} \tag{7}$$

To obtain the force tracking error in the steady state, the system can be brought to a steady state. When $s \to 0$, all differential terms of the system are equal to zero, and the steady-state force tracking error is given by:

$$\Delta f_{ss} = \lim_{s\to 0} s\Delta f(s) = \frac{k_r}{k_r + k_e}\left[k_e\left(z - z_d\right) - f_{Nd}\right] \tag{8}$$

The above equation indicates that for the system to achieve asymptotic stability, it is necessary to satisfy $z_d = f_{Nd}/k_e$ or $k_r = 0$. However, due to the complex time-varying nature of environmental stiffness during complex surface tracking, obtaining an accurate k_e is difficult. Therefore, choosing $k_r = 0$ can satisfy the condition for asymptotic stability, and the simplified impedance equation (IC) is obtained as follows Sheng and Zhang (2018):

$$e(t) = m_r\Delta\ddot{z}(t) + b_r\Delta\dot{z}(t) \tag{9}$$

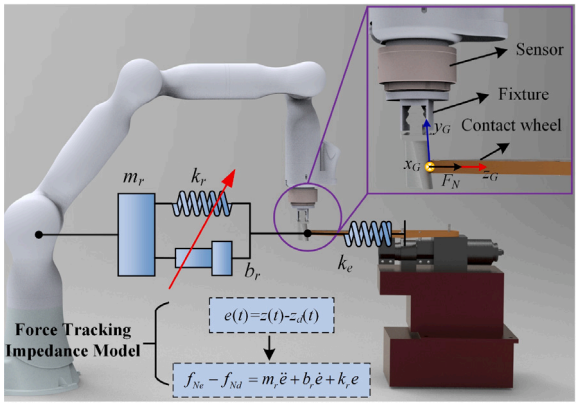


Fig. 4. Impedance control strategy is adopted for robot environment interaction.

3.2. Dynamic adaptive impedance control

The dynamic adaptive impedance control mainly consists of two components: the dynamic observer-based adaptive impedance controller (Section 3.2.1) and the rolling parameter optimization (Section 3.2.2), where the rolling parameter optimization is used to optimize the key parameters within the dynamic observer.

3.2.1. Dynamic observer-based adaptive impedance controller

Due to the dynamic and complex changes of the contact environment, traditional impedance controllers exhibit poor robustness when dealing with nonlinear systems. An adaptive impedance controller (AIC) is introduced in Duan et al. (2018) to enhance force tracking performance in complex contact environments:

$$\begin{cases} e(t) = m_r\Delta\ddot{z}(t) + b_r(\Delta\dot{z}(t) + \rho(t)) \\ \rho(t) = \rho(t-T) + \phi\frac{c(t-T)}{b_r} \\ c(t-T) = f_{Nd}(t-T) - f_{Ne}(t-T) \end{cases} \tag{10}$$

where T is the sampling period, $\rho(t)$ is the adaptive compensation term at time t, ϕ is the update rate, and $c(t-T)$ represents the contact force error value of the previous cycle. Compared to the IC, the AIC adopts an adaptive mechanism that dynamically adjusts the contact force error during the contact process to reduce the occurrence of contact force errors.

From the design of the adaptive mechanism, it can be observed that a larger ϕ improves the force tracking accuracy; however, it may also cause system oscillations. Conversely, if ϕ is too small, the force tracking accuracy becomes insufficient. Therefore, keeping ϕ constant is not the optimal control solution. To address this, the dynamic observer (DO) is designed to adjust ϕ in real time, and its expression is defined as:

$$\phi = \frac{1}{e^{\omega|c(t-T)|} + e^{\tau|c'(t-T)|} + B_{obs}} \tag{11}$$

The purpose of the dynamic observer design is to make the update rate ϕ dynamically adjustable. In the dynamic observer, ω represents the weighting coefficient of the contact force variation, τ represents the weighting coefficient of the contact force variation rate, and B_{obs} serves as the stability coefficient to ensure the system remains in an absolutely stable state. Due to the design structure of the dynamic observer, both ω and τ can influence the variation trend of ϕ.

According to Eq. (10), the error transfer function of the AIC can be expressed as:

$$\begin{cases} m_r\ddot{\eta}(t) + b_r\dot{\eta}(t) + \phi k_e\eta(t-T) + k_e\eta(t) \\ \quad + b_r k_e\rho(t-T) = m_r\ddot{\gamma}(t) + b_r\dot{\gamma}(t) \\ \eta(t) = f_{Nd}(t) - f_{Ne}(t) \\ \gamma(t) = f_{Nd}(t) - \hat{f}_{Ne}(t) \end{cases} \tag{12}$$

Here, $\hat{f}_{Ne}(t)$ represents the estimated contact force error in the environment, and the adaptive compensation term $b_r\rho(t-T)$ from the previous period is expanded as follows:

$$b_r\rho(t-T) = \phi\eta(t-(n+1)T) + \cdots + \phi\eta(t-2T) \tag{13}$$

Substituting Eq. (13) into Eq. (12), the error transfer function equation is obtained as:

$$\begin{aligned} &m_r\ddot{\eta}(t) + b_r\dot{\eta}(t) + \phi k_e\eta(t) + k_e\eta(t) + \phi k_e \\ &(\eta(t-(n+1)T) + \cdots + \phi\eta(t-T)) = m_r\ddot{\gamma}(t) + b_r\dot{\gamma}(t) \end{aligned} \tag{14}$$

Applying the Laplace transform to the equation yields the error transfer function as follows:

$$\frac{\eta(s)}{\gamma(s)} = \frac{m_r s^2 + b_r s}{m_r s^2 + b_r s + k_e + \phi k_e\left(e^{-(n+1)Ts} + \cdots + e^{-Ts}\right)} \tag{15}$$

The stability of the error transfer function is guaranteed by its characteristic equation:

$$m_r s^2 + b_r s + k_e + \phi k_e\left(e^{-(n+1)Ts} + \cdots + e^{-Ts}\right) = 0 \tag{16}$$

When n is sufficiently large and the sampling rate is adequate ($T \ll 1s$), it can be approximately assumed that $e^{-Ts} \cong 1 - Ts$. Under these conditions, the characteristic equation is rewritten as:

$$m_r T s^3 + b_r T s^2 + k_e T(1-\phi)s + \phi k_e = 0 \tag{17}$$

For the characteristic equation, the stability conditions are obtained based on the Routh criterion:

$$\begin{cases} \frac{b_r k_e T^2(1-\phi) - m_r k_e T\phi}{b_r T} > 0 \\ k_e T(1-\phi) > 0 \\ \phi k_e > 0 \end{cases} \tag{18}$$

The boundary condition of ϕ is given as:

$$0 < \phi < \frac{b_r T}{m_r + b_r T} \tag{19}$$

Substituting Eq. (11) into the above equation, the stability condition for DO-AIC is obtained as:

$$B_{obs} > \frac{m_r + b_r T}{b_r T} - \left(e^{\sigma|\Delta f|} + e^{\tau|\Delta f'|}\right) \tag{20}$$

For a stable system, the steady-state error e_{ss} can be defined through the Laplace transform as:

$$\begin{aligned} e_{ss} &= \lim_{t\to\infty}\eta(t) = \lim_{s\to 0} s(\eta(s) - \ell(s)) \\ &= \lim_{s\to 0} s\left(\frac{m_r s^2 + b_r s}{m_r s^2 + b_r s + k_e + \phi k_e\left(e^{-(n+1)Ts} + \cdots + e^{-Ts}\right)}\ell(s) - \ell(s)\right) \end{aligned} \tag{21}$$

where $\eta(s)$ and $\ell(s)$ represent the output and input signals of the system in the frequency domain, respectively. Assuming the input of the system is a step response function, namely $\ell(s) = 1/s$, and the steady-state error of the system can be obtained based on the above equation as:

$$e_{ss} = \lim_{s\to 0} s(\eta(s) - \ell(s)) = -1 \tag{22}$$

Based on the calculation results, the conclusion can be drawn as:

$$\lim_{s\to 0} s\eta(s) = 0 \to \lim_{t\to\infty}\eta(t) = f_{Nd}(t) - f_{Ne}(t) = 0 \tag{23}$$

It is proven that as time increases, the contact force error will eventually approach zero, ensuring the asymptotic stability of the DO-AIC controller. The stability condition requires that the stability coefficient B_{obs} satisfies the stability condition Eq. (20).

3.2.2. Rolling parameter optimization

In the dynamic adaptive impedance controller, the two weighting coefficients of the dynamic observer remain constant, which makes it difficult to ensure good contact force control performance. Accordingly, inspired by the rolling estimation strategy in moving horizon estimation, the two weighting coefficients are optimized within a sliding time window based on historical data. This process is herein referred to as "rolling parameter optimization". In each control cycle, the optimal parameters are determined by solving a quadratic programming problem, enabling real-time optimization of the dynamic observer to address complex contact environments.

Based on Eqs. (10) and (11), the state update expressions for displacement and velocity can be obtained as:

$$\begin{cases} z(t+T) = z(t) + \dot{z}(t)T \\ \dot{z}(t+T) = \dot{z}(t) + \left(\frac{e(t) - b_r\left(\dot{z}(t) - \dot{z}_d - \rho(t)\right)}{m_r} + \ddot{z}_d\right)T \end{cases} \tag{24}$$

To ensure the minimization of steady-state error while suppressing contact overshoot, the following cost function is constructed:

$$\begin{aligned} \min_{\omega,\tau} J = {} & w_1 \sum_{i=t-\left(N_m-1\right)T}^{t} e(i)^2 \\ & + w_2\left(\max_{t-\left(N_m-1\right)T\le i\le t} |\, e(i)\right)^2 + \lambda_\omega\omega^2 + \lambda_\tau\tau^2 \end{aligned} \tag{25}$$

Here, N_m is the length of the optimization horizon, w_1 and w_2 are the weights for steady-state error and overshoot, respectively; λ_ω and λ_τ are the regularization parameters. ω^2 and τ^2 are the regularization terms.

Since the dynamic observer is designed in an exponential form, the influence of the key parameters ω and τ on the contact force error cannot be obtained through a linear approach. Therefore, the gradient descent method is used to determine the sensitivity of the contact force error to the two key parameters. Using this gradient information, quadratic optimization is performed at each time step to obtain the optimal values of ω and τ. Here, ω and τ influence the contact force error by changing the update rate ϕ. Therefore, it is necessary to first calculate the partial derivatives of $\phi(t)$ with respect to ω and τ, which are given as:

$$\begin{cases} \frac{\partial\phi(t)}{\partial\omega} = -\frac{e^{(\omega|c(t-T)|)}|c(t-T)|}{\left(e^{(\omega|c(t-T)|)} + e^{\left(\tau|c'(t-T)|\right)} + B_{obs}\right)^2} \\ \frac{\partial\phi(t)}{\partial\tau} = -\frac{e^{\left(\tau|c'(t-T)|\right)}|c'(t-T)|}{\left(e^{(\omega c(t-T))} + e^{\left(\tau|c'(t-T)|\right)} + B_{obs}\right)^2} \end{cases} \tag{26}$$

Since $\rho(t)$ is an accumulated quantity, the gradients with respect to ω and τ are:

$$\begin{cases} \frac{\partial\rho(t)}{\partial\omega} = \frac{\partial\rho(t-T)}{\partial\omega} + \frac{1}{b_r}\frac{\partial\phi(t)}{\partial\omega}c(t-T) \\ \frac{\partial\rho(t)}{\partial\tau} = \frac{\partial\rho(t-T)}{\partial\tau} + \frac{1}{b_r}\frac{\partial\phi(t)}{\partial\tau}c(t-T) \end{cases} \tag{27}$$

The partial derivatives of $e(t)$ with respect to ω and τ can be calculated as:

$$\begin{cases} \frac{\partial e(t)}{\partial\omega} = b_r\frac{\partial\rho(t)}{\partial\omega} \\ \frac{\partial e(t)}{\partial\tau} = b_r\frac{\partial\rho(t)}{\partial\tau} \end{cases} \tag{28}$$

Let $\chi = \max_{t-\left(N_m-1\right)T\le i\le t}|e(i)|$ represents the maximum absolute contact force error up to the current time. Its partial derivatives with respect to ω and τ are:

$$\begin{aligned} \frac{\partial\chi}{\partial\omega} &= \begin{cases} \operatorname{sgn}(e(t^*))\frac{\partial e(t^*)}{\partial\omega}, & \text{if } |e(t^*)| = \chi \\ 0, & \text{else} \end{cases} \\ \frac{\partial\chi}{\partial\tau} &= \begin{cases} \operatorname{sgn}(e(t^*))\frac{\partial e(t^*)}{\partial\tau}, & \text{if } |e(t^*)| = \chi \\ 0, & \text{else} \end{cases} \end{aligned} \tag{29}$$

The partial derivatives of the objective function J with respect to ω and τ can be calculated as:

$$\begin{cases} \frac{\partial J}{\partial\omega} = 2w_1 e(t)\frac{\partial e(t)}{\partial\omega} + 2w_2\chi\frac{\partial\chi}{\partial\omega} + 2\lambda_\omega\omega \\ \frac{\partial J}{\partial\tau} = 2w_1 e(t)\frac{\partial e(t)}{\partial\tau} + 2w_2\chi\frac{\partial\chi}{\partial\tau} + 2\lambda_\tau\tau \end{cases} \tag{30}$$

The optimization problem is solved using quadratic programming, converting the problem into a standard quadratic programming form:

$$\begin{cases} \min\limits_{\omega,\tau} \frac{1}{2}\boldsymbol{\kappa}^T \boldsymbol{H}\boldsymbol{\kappa} + \boldsymbol{f}^T\boldsymbol{\kappa} \\ \text{subject to} \ : u_\omega \le \omega \le l_\omega, u_\tau \le \tau \le l_\tau \end{cases} \tag{31}$$

Here, $\boldsymbol{\kappa} = [\omega, \tau]^T$, where the gradient of the objective function concerning the optimization variables is expressed as $\boldsymbol{f} = \left[\frac{\partial J}{\partial \omega}, \frac{\partial J}{\partial \tau}\right]^T$. $\boldsymbol{H}$ represents the Hessian matrix, which is a diagonal matrix with entries $2\lambda_\omega$ and $2\lambda_\tau$. u_ω and l_ω represent the upper and lower bound for ω, and similarly for τ. Finally, the entire quadratic programming problem can be expressed as:

$$\begin{cases} \min\limits_{\omega,\tau} \frac{1}{2}\begin{bmatrix} \omega \\ \tau \end{bmatrix}^T 2\begin{bmatrix} \lambda_\omega & 0 \\ 0 & \lambda_\tau \end{bmatrix}\begin{bmatrix} \omega \\ \tau \end{bmatrix} + \begin{bmatrix} \frac{\partial J}{\partial \omega} \\ \frac{\partial J}{\partial \tau} \end{bmatrix}^T \begin{bmatrix} \omega \\ \tau \end{bmatrix} \\ \text{subject to} : u_\omega \le \omega \le l_\omega, u_\tau \le \tau \le l_\tau \end{cases} \tag{32}$$

Solving the above equation gives the optimized $\boldsymbol{\kappa}^* = [\omega^*, \tau^*]^T$, with ω^* and τ^* as the inputs to the current dynamic observer. This allows for the improvement of the overall performance of DO-AIC.

3.3. Model predictive dynamic adaptive impedance control algorithm

The state-space vector of model predictive control is:

$$\boldsymbol{\xi} = \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix} = \begin{bmatrix} z \\ \dot{z} \\ f_{Ne} \end{bmatrix} \tag{33}$$

Assume that the input to the controller is the velocity difference in Cartesian space $u = \dot{z} - \dot{z}_d = \Delta\dot{z}$, The desired position z_d is generated by the dynamic adaptive impedance model, and its first-order derivative $\dot{z}_d$ is estimated in the discrete-time system using finite difference approximation, thereby providing a well-defined reference velocity input for the controller. The state vector $\boldsymbol{\xi}$ is treated as the optimization variable in the cost function.

The state-space vector is discretized using the forward Euler method, and the resulting expression is given by:

$$\boldsymbol{\xi}(k+1) = \begin{bmatrix} \xi_1(k) + T\dot{\xi}_1(k) \\ \xi_2(k) + T\dot{\xi}_2(k) \\ f_{Ne}(k) + T\dot{f}_{Ne}(k) \end{bmatrix} \tag{34}$$

Specifically, the components of the state-space vector can be expressed as:

$$\begin{cases} \dot{\xi}_1 = \xi_2 \\ \dot{\xi}_2 = \left[\ddot{z}_d + \frac{1}{m_r}\left(f_{Ne} - f_{Nd} - b_r\left(\dot{z} - \dot{z}_d - \rho\right)\right)\right] \\ \dot{f}_{Ne} = k_e\left(\dot{z} - \dot{z}_d\right) \end{cases} \tag{35}$$

Express it in the form of a state-space equation:

$$\xi(k+1) = \boldsymbol{\Gamma}\xi(k) + \boldsymbol{\Sigma}u(k) + \boldsymbol{Q} \tag{36}$$

In Eq. (36), $\boldsymbol{\Gamma}$, $\boldsymbol{\Sigma}$, and $\boldsymbol{Q}$ can be specifically expressed as:

$$\begin{aligned} \boldsymbol{\Gamma} &= \begin{bmatrix} 1 & T & 0 \\ 0 & 1 & T/m_r \\ 0 & b_r + Tk_e & 0 \end{bmatrix}, \\ \boldsymbol{\Sigma} &= \begin{bmatrix} 0 \\ \frac{-Tb_r}{m_r} \\ b_r \end{bmatrix}, \boldsymbol{Q} = \begin{bmatrix} 0 \\ \frac{-T}{m_r} f_{Nd} \\ f_{Nd} \end{bmatrix} \end{aligned} \tag{37}$$

Let the prediction horizon be N_n, and define the symbol $\xi\left(k+N_n \mid k\right)$ to represent the value of the variable ξ predicted at time $k+N_n$. Then:

$$\begin{cases} \xi(k \mid k) = \xi_k \\ \xi(k+1 \mid k) = \boldsymbol{\Gamma}\xi(k \mid k) + \boldsymbol{\Sigma}u(k \mid k) + \boldsymbol{Q} \\ \qquad\qquad = \boldsymbol{\Gamma}\xi_k + \boldsymbol{\Sigma}u(k \mid k) + \boldsymbol{Q} \\ \xi(k+2 \mid k) = \boldsymbol{\Gamma}\xi(k+1 \mid k) + \boldsymbol{\Sigma}u(k+1 \mid k) \\ \qquad\qquad + \boldsymbol{Q} = \boldsymbol{\Gamma}^2\xi_k + \boldsymbol{\Gamma}\boldsymbol{\Sigma}u(k \mid k) \\ \qquad\qquad + \boldsymbol{\Sigma}u(k+1 \mid k) + (\boldsymbol{\Gamma} + \boldsymbol{I})\boldsymbol{Q} \\ \qquad\qquad \vdots \\ \xi\left(k+N_n \mid k\right) = \boldsymbol{\Gamma}^{N_n}\xi_k + \boldsymbol{\Gamma}^{N_n-1}\boldsymbol{\Sigma}u(k \mid k) \\ \qquad\qquad + \cdots + \boldsymbol{\Sigma}u\left(k+N_n-1 \mid k\right) \\ \qquad\qquad + \left(\boldsymbol{\Gamma}^{N_n-1} + \boldsymbol{\Gamma}^{N_n-2} + \cdots + \boldsymbol{I}\right)\boldsymbol{Q} \end{cases} \tag{38}$$

Define:

$$\begin{aligned} \boldsymbol{\Xi}_k &= \begin{bmatrix} \xi(k \mid k) \\ \xi(k+1 \mid k) \\ \vdots \\ \xi\left(k+N_n \mid k\right) \end{bmatrix}, \\ \boldsymbol{U}_k &= \begin{bmatrix} u(k \mid k) \\ u(k+1 \mid k) \\ \vdots \\ u\left(k+N_n-1 \mid k\right) \end{bmatrix} \end{aligned} \tag{39}$$

Eq. (38) can be written as $\boldsymbol{\Xi}_k = \boldsymbol{L}\xi_k + \boldsymbol{K}\boldsymbol{U}_k + \boldsymbol{Q}_p$.

Where $\boldsymbol{L}, \boldsymbol{K}, \boldsymbol{Q}_p$ can be written as:

$$\begin{aligned} \boldsymbol{L} &= \begin{bmatrix} \boldsymbol{I} \\ \boldsymbol{\Gamma} \\ \vdots \\ \boldsymbol{\Gamma}^{N_n} \end{bmatrix}, \boldsymbol{K} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ \boldsymbol{\Sigma} & 0 & \cdots & 0 \\ \boldsymbol{\Gamma}\boldsymbol{\Sigma} & \boldsymbol{\Sigma} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{\Gamma}^{N_n-1}\boldsymbol{\Sigma} & \boldsymbol{\Gamma}^{N_n-2}\boldsymbol{\Sigma} & \cdots & \boldsymbol{\Sigma} \end{bmatrix}, \\ \boldsymbol{Q}_p &= \begin{bmatrix} 0 \\ \boldsymbol{I} \\ \boldsymbol{\Gamma} + \boldsymbol{I} \\ \vdots \\ \frac{\boldsymbol{I} - \boldsymbol{\Gamma}^{N_n}}{\boldsymbol{I} - \boldsymbol{\Gamma}} \end{bmatrix}\boldsymbol{Q} \end{aligned} \tag{40}$$

The above equation is used as the predictive model within the prediction horizon N_n, where $\boldsymbol{L}$, $\boldsymbol{K}$, and $\boldsymbol{Q}_p$ are all calculated based on the current system state and remain constant within the prediction horizon. Through the rolling optimization strategy, the system state variable $\boldsymbol{\Xi}_{k+1}$ at the next time step is computed.

Since the loss function needs to optimize the variation of the input vector, let $\boldsymbol{E} = \boldsymbol{\xi} - \boldsymbol{\xi}_r$, where $\boldsymbol{\xi}_r$ is the reference input and $\boldsymbol{\xi}$ is the actual input. The performance metric is to minimize the error, which approaches zero, and the following cost function is established:

$$\begin{aligned} &\min \Omega \\ &= \sum_{i=0}^{N_n-1} [\boldsymbol{E}(k+i \mid k)^T \boldsymbol{W}_{\xi}\boldsymbol{E}(k+i \mid k) \\ &+ u(k+i \mid k)^T W_u u(k+i \mid k)] \\ &+ \boldsymbol{E}\left(k+N_n\right)^T \boldsymbol{W}_f \boldsymbol{E}\left(k+N_n\right) \\ &= \sum_{i=0}^{N_n-1} \{\left[\boldsymbol{\xi}(k+i \mid k) - \boldsymbol{\xi}_r(k+i \mid k)\right]^T \\ &\boldsymbol{W}_{\xi}\left[\boldsymbol{\xi}(k+i \mid k) - \boldsymbol{\xi}_r(k+i \mid k)\right] \\ &+ u(k+i \mid k)^T W_u u(k+i \mid k)\} \\ &+ \left[\boldsymbol{\xi}\left(k+N_n \mid k\right) - \boldsymbol{\xi}_r\left(k+N_n \mid k\right)\right]^T \\ &\boldsymbol{W}_f\left[\boldsymbol{\xi}\left(k+N_n \mid k\right) - \boldsymbol{\xi}_r\left(k+N_n \mid k\right)\right] \end{aligned} \tag{41}$$

where $\boldsymbol{W}_{\xi}, \boldsymbol{W}_f$ represent the error weight matrix, and terminal error weight matrix, respectively. The W_u is the input weight coefficient. The

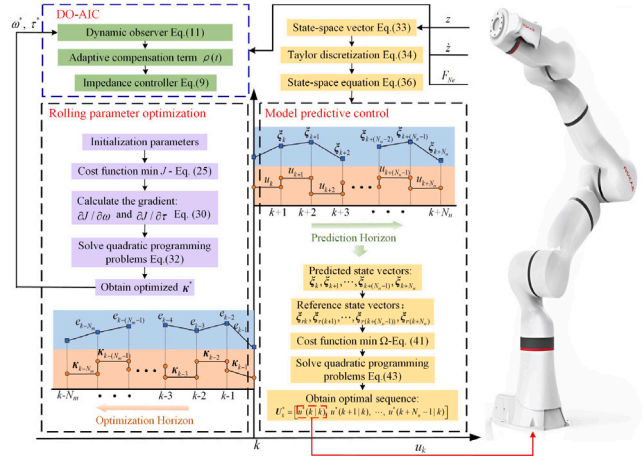


Fig. 5. Control algorithm flow of MPRPO-DAIC.

expression of Eq. (41) in matrix form and its expansion are as follows:

$$
\begin{aligned}
&\min \Omega \\
&= \begin{bmatrix} \xi(k \mid k) \\ \xi(k+1 \mid k) \\ \vdots \\ \xi\left(k+N_n \mid k\right) \end{bmatrix}^T \begin{bmatrix} \boldsymbol{W}_\xi & 0 & 0 \\ 0 & \boldsymbol{W}_\xi & 0 \\ \vdots & \ddots & \vdots \\ 0 & \vdots & \boldsymbol{W}_f \end{bmatrix} \begin{bmatrix} \xi(k \mid k) \\ \xi(k+1 \mid k) \\ \vdots \\ \xi\left(k+N_n \mid k\right) \end{bmatrix} \\
&- 2 \begin{bmatrix} \xi(k \mid k) \\ \xi(k+1 \mid k) \\ \vdots \\ \xi\left(k+N_n \mid k\right) \end{bmatrix}^T \begin{bmatrix} \boldsymbol{W}_\xi & 0 & 0 \\ 0 & \boldsymbol{W}_\xi & 0 \\ \vdots & \ddots & \vdots \\ 0 & \vdots & \boldsymbol{W}_f \end{bmatrix} \begin{bmatrix} \xi_r(k \mid k) \\ \xi_r(k+1 \mid k) \\ \vdots \\ \xi_r\left(k+N_n \mid k\right) \end{bmatrix} \\
&+ \begin{bmatrix} \xi_r(k \mid k) \\ \xi_r(k+1 \mid k) \\ \vdots\ \xi_r\left(k+N_n \mid k\right) \end{bmatrix}^T \begin{bmatrix} \boldsymbol{W}_\xi & 0 & 0 \\ 0 & \boldsymbol{W}_\xi & 0 \\ \vdots & \ddots & \vdots \\ 0 & \vdots & \boldsymbol{W}_f \end{bmatrix} \begin{bmatrix} \xi_r(k \mid k) \\ \xi_r(k+1 \mid k) \\ \vdots \\ \xi_r\left(k+N_n \mid k\right) \end{bmatrix} \\
&+ \begin{bmatrix} u(k \mid k) \\ u(k+1 \mid k) \\ \vdots \\ u\left(k+N_n-1 \mid k\right) \end{bmatrix}^T W_u \begin{bmatrix} u(k \mid k) \\ u(k+1 \mid k) \\ \vdots \\ u\left(k+N_n-1 \mid k\right) \end{bmatrix} \\
&= \Xi_k^T \overline{\boldsymbol{W}}_\xi \Xi_k - 2\Xi_k^T \overline{\boldsymbol{W}}_\xi \Xi_{rk} + \Xi_{rk}^T \overline{\boldsymbol{W}}_\xi \Xi_{rk} + \boldsymbol{U}_k^T W_u \boldsymbol{U}_k
\end{aligned} \tag{42}
$$

Substituting $\Xi_k = \boldsymbol{L}\xi_k + \boldsymbol{K}\boldsymbol{U}_k + \boldsymbol{Q}_p$ into the above equation and writing it in the form of a quadratic programming problem:

$$
\begin{cases}
\boldsymbol{U}_k^* = \min_{U_k} [\xi_k^T \boldsymbol{L}^T \overline{\boldsymbol{W}}_\xi \boldsymbol{L}\xi_k + 2\boldsymbol{U}_k^T (\boldsymbol{K}^T \overline{\boldsymbol{W}}_\xi \boldsymbol{L}\Xi_k \\
\quad + \boldsymbol{K}^T \overline{\boldsymbol{W}}_\xi \boldsymbol{Q}_p - \boldsymbol{K}^T \overline{\boldsymbol{W}}_\xi \Xi_{rk}) \\
\quad + \boldsymbol{U}_k^T (\boldsymbol{K}^T \overline{\boldsymbol{W}}_\xi \boldsymbol{K} + W_u) \boldsymbol{U}_k] \\
\text{subject to} \ : \upsilon_{\min} \le u_k \le \upsilon_{\max}
\end{cases} \tag{43}
$$

The optimal sequence obtained at each prediction horizon is solved, and only the first value is applied to the controlled object, i.e., $u_k = u^*(k \mid k) = \left[\boldsymbol{I}_{p\times p}, 0, 0, \ldots, 0\right] \boldsymbol{U}_k^*$, where u_k is the predicted velocity difference at the kth time step. It is then integrated to compensate for the robotic end-effector position at the $k+1$-th time step. The entire control process of MPRPO-DAIC is shown in Fig. 5.

4. Simulation experiment

To verify the advancement of MPRPO-DAIC, Impedance Controller (IC), Adaptive Impedance Controller (AIC), Dynamic Observer-based Adaptive Impedance Controller (DO-AIC), and Rolling Parameter Optimization-based Dynamic Adaptive Impedance Controller (RPO-DAIC) are introduced as comparison controllers. The AIC is proposed in Duan et al. (2018). The differences between five controllers are shown in Table 1. The simulation models for different controllers are built using Simulink 10.2/R2020a, with the basic impedance controller parameters set as $m_r = 1\ \mathrm{Ns^2/m}$ and $b_r = 200\ \mathrm{Ns/m}$, and the sampling period $T = 4$ ms. The desired tracking force is set to $f_{Nd} = 10$ N. The simulation model of MPRPO-DAIC is built as shown in Fig. 6.

Table 1
Comparison of differences across different controllers.

Controller type	IC	AIC	DO-AIC	RPO-DAIC	MPRPO-DAIC
Adaptive compensation term	N	Y	Y	Y	Y
Dynamic observer	N	N	Y	Y	Y
Rolling parameter optimization	N	N	N	Y	Y
Model predictive control	N	N	N	N	Y

To verify the effectiveness of dynamic observer, simulations are conducted based on IC, AIC, and DO-AIC, where the weight coefficients ω and τ in DO-AIC are both set to 1. Compared to DO-AIC, the ϕ in AIC is a fixed value that adhered to boundary conditions. Larger and smaller values are chosen within the boundary conditions, set to 0.01 and 0.1, respectively. When ϕ in AIC is set to 0, the adaptive compensation term is absent, and AIC becomes IC. To study the force tracking performance of three controllers in complex environments, the environment stiffness is set to $k_e = 5000 + 3000\sin(\pi/2t)$ and $k_e = 5000 + 300\sin(2\pi t)$, representing large deformation and high frequency changing environment, respectively.

The simulation results are shown in Fig. 7. During the initial contact phase ($0-1s$), as ϕ increases in AIC, the system exhibits larger vibrations and contact force overshoot. Compared to AIC, both IC and DO-AIC have stronger abilities to suppress system vibrations, showing this trend in both large deformation and high frequency changing environments. During the steady-state tracking phase (1–10 s), for the large deformation environment, the larger the ϕ, the smaller the force tracking error, resulting in higher force tracking accuracy. In the high frequency changing environment, it can be observed that both IC and AIC perform poorly in adapting to high frequency changes. For AIC, as ϕ increases, the force tracking error decreases. DO-AIC still shows the best steady-state contact force tracking ability. It can be seen that, due to the presence of the dynamic observer, DO-AIC can simultaneously ensure strong suppression of system vibrations, as well as a small steady-state contact force error when facing complex contact environments.

As depicted in Fig. 8(a), (b), (c), and (d), whether in the large deformation or high frequency changing environment, the adaptive compensation term in AIC undergoes significant changes during the initial contact phase, leading to large contact force errors. The larger the ϕ, the more pronounced this phenomenon becomes. If ϕ is too small, it can result in insufficient compensation, causing excessive contact force tracking errors during the steady-state tracking phase. Due to the presence of the dynamic observer, DO-AIC can dynamically adjust the update rate ϕ, maintaining a smaller ϕ during the initial contact phase to suppress system vibrations, and a larger ϕ during the steady-state contact phase to improve force control accuracy. The force error curves shown in Fig. 8(b) and (d) illustrate that the DO-AIC achieves rapid error decay and maintains a steady-state value close to zero within approximately 0.25 s in both conditions. Compared to AIC ($\phi = 0.01$) and AIC ($\phi = 0.1$), which either exhibit longer settling time or persistent oscillations, DO-AIC demonstrates superior convergence speed and robustness. These results provide quantitative support for the theoretical conclusion and confirm the practical feasibility of the DO-AIC control framework in dynamic and uncertain contact environments.

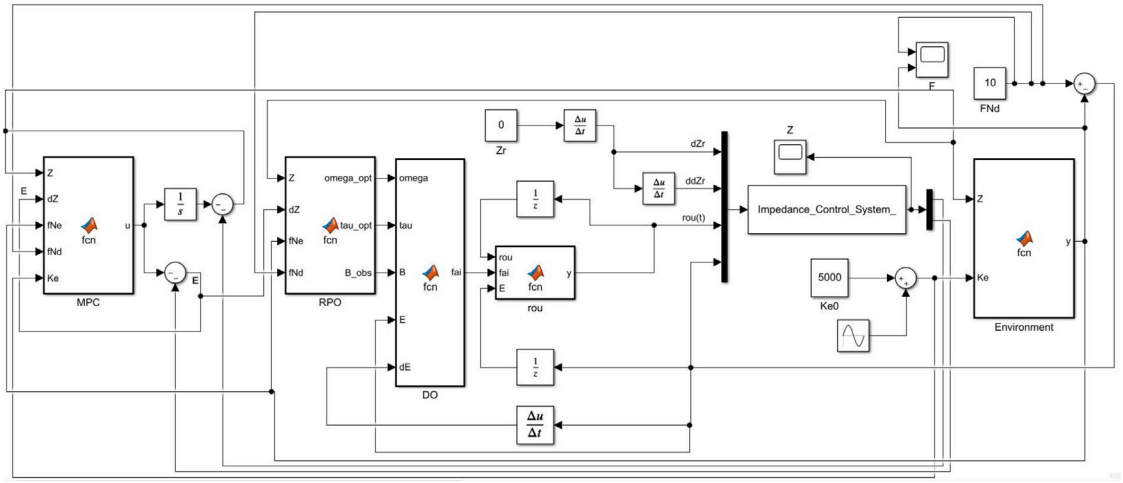
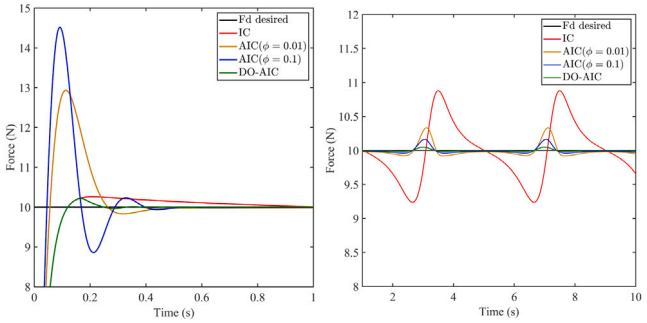


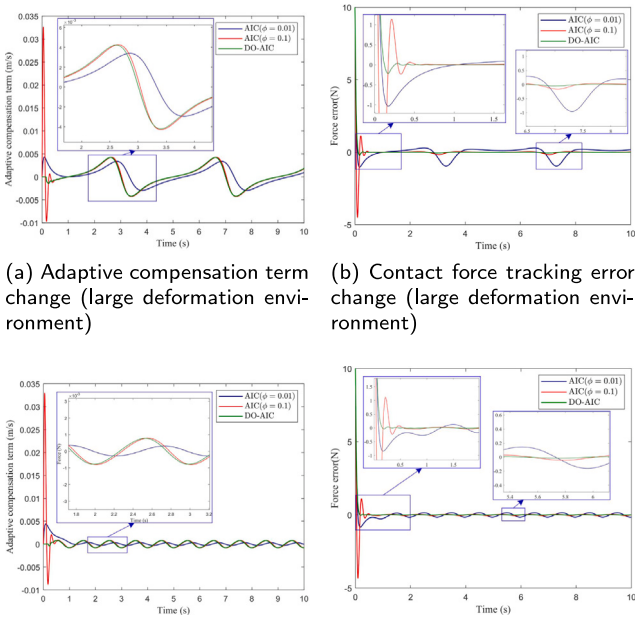
Fig. 6. Simulink simulation model of MPRPO-DAIC.



(a) Large deformation environment

(b) High frequency changing environment

Fig. 7. Force tracking performance of DO-AIC, AIC, and IC under the complex environment.

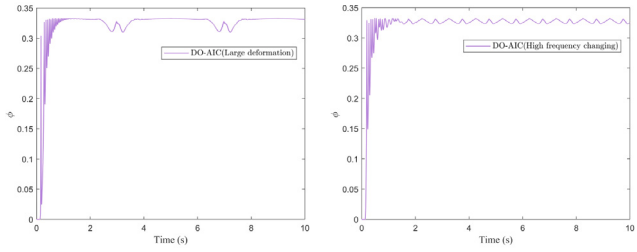


(a) Adaptive compensation term change (large deformation environment)

(b) Contact force tracking error change (large deformation environment)

(c) Adaptive compensation term change (high-frequency changing environment)

(d) Contact force tracking error change (high-frequency changing environment)



(e) update rate change (large deformation contact environment)

(f) update rate change (high-frequency changing environment)

Fig. 8. Comparison of key variables in AIC and DO-AIC under the complex environment (adaptive compensation term, contact force error, update rate).

As illustrated in Fig. 8(e) and (f), for the large deformation environment, significant contact force errors occur at areas of large deformation, and the ϕ value can decrease accordingly to maintain system stability. For the high frequency changing environment, ϕ changes at a frequency corresponding to the environmental changes. Based on the simulation results presented above, it is evident that the presence of the dynamic observer combines the advantages of IC and AIC in force tracking, ensuring stable suppression of contact force during the initial contact phase while also maintaining force tracking accuracy during the steady-state contact phase, thus improving the robustness of AIC in complex contact environments.

Fig. 9 compares the root mean square error (RMSE) of force tracking error under large deformation and high-frequency changing environments for three control strategies. In both cases, DO-AIC achieves the

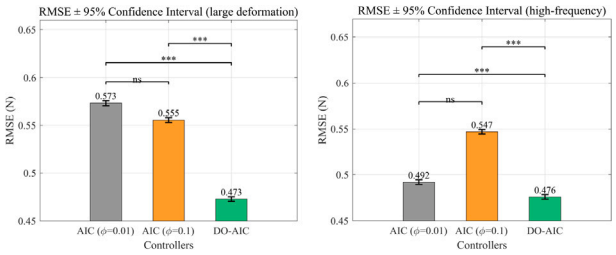
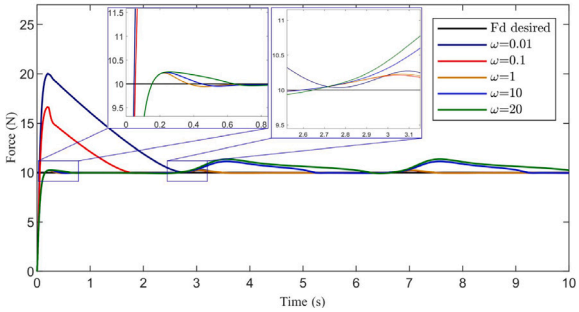
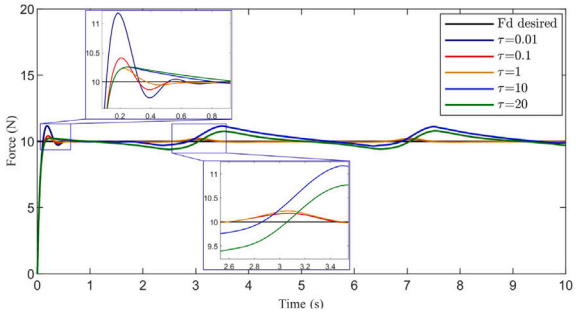


Fig. 9. Comparison of force error RMSE and statistical significance under different environments.



(a) Controller performance under different ω ($\tau = 1$)



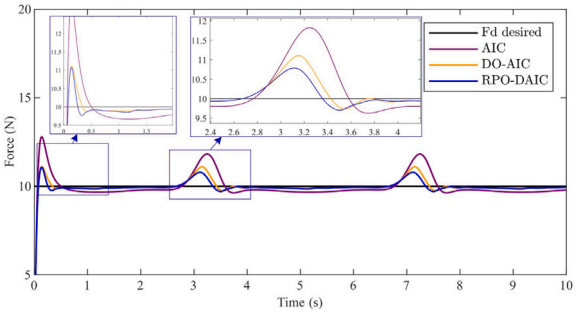
(b) Controller performance under different $\tau(\omega = 1)$

Fig. 10. The Influence of weight coefficients in DO-AIC on controller performance.

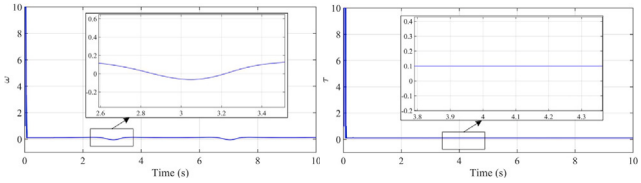
lowest RMSE, with values of 0.473 N in the large deformation scenario and 0.476 N in the high-frequency scenario. The statistical analysis indicates that the improvements achieved by DO-AIC over AIC (ϕ = 0.01) and AIC (ϕ = 0.1) are highly significant, confirming that the observed differences are not due to random fluctuations. In contrast, the performance difference between the two AIC configurations is not statistically significant. These results highlight the superior robustness and force control accuracy of DO-AIC under complex and dynamic contact environments.

For DO-AIC, the weight coefficients ω and τ directly affect the sensitivity of the controller to contact force errors and their rate of change, thus directly influencing the performance of the controller.

Therefore, simulations are conducted on the key parameters ω and τ under large deformation stiffness, with the stability coefficient B_{obs} set to 1. As shown in Fig. 10, decreasing ω and τ causes the system to exhibit force overshoot. This is because smaller weight coefficients reduce the sensitivity of system to contact force information. Conversely, as the weight coefficients increase, the update rate becomes increasingly sensitive to the feedback information. This also causes the update rate to fluctuate over a wider range, thereby reducing the stability and robustness of the controller, making precise contact force tracking more difficult. Therefore, parameter optimization for the weight coefficients ω and τ is necessary.



(a) Comparison of experimental results for three controllers



(b) Variation of parameters ω and τ

Fig. 11. Comparison of the three controllers and the variation of parameters ω and τ during the experiments.

To better analyze the variations of the two key parameters, ω and τ, during the rolling parameter optimization process, simulations were conducted based on AIC, DO-AIC, and RPO-DAIC. In DO-AIC, both ω and τ were set to 1, while in RPO-DAIC, the optimization horizon N_m for the parameter optimization was set to 10. The simulation results are shown in Fig. 11(a), and the specific changes of ω and τ during the parameter optimization are shown in Fig. 11(b). It can be observed that at the initial contact stage, both ω and τ quickly increase to 10 to achieve rapid force convergence at the start of contact. However, as the contact oscillations decrease, ω and τ gradually decrease to improve the compensatory effect of the controller, thereby reducing steady-state contact errors. During steady-state contact, ω varies with the force error in a periodic manner, where ω decreases as the force error increases. During steady-state contact, τ remains relatively constant because the rate of change of the force error is much smaller than that of the force error itself, so τ stabilizes around 0.1.

To evaluate the performance of the proposed MPRPO-DAIC, force tracking simulation experiments are conducted for five different controllers in various contact scenarios. The simulations include three types of contact environments: plane, large deformation environment, and high frequency changing environment. The force control accuracy is evaluated by calculating RMSE of force signals (Mu et al., 2023). The optimization horizon N_m for the parameter optimization was set to 10. Since the optimization problem is formulated as a low-dimensional convex quadratic program, the average execution time of the optimization module is approximately 1.2 ms, measured on a standard PC platform (Intel Core i9-10900K, 3.7 GHz, 32 GB RAM), which is considered acceptable. The specific parameters of MPRPO-DAIC are listed in Table 2, where the prediction horizon N_n is set to 5, and the corresponding computation time of the MPC part is approximately 13 ms, which is considered acceptable for robotic grinding applications (Wang et al., 2023a).

Fig. 12 illustrates the force tracking performance during plane tracking, with the environment stiffness set to $k_e = 5000 + 600\sin(2\pi t)$. Fig. 15 indicates that the force control accuracy of MPRPO-DAIC is 0.258 N, exhibiting the best force tracking performance among the five controllers. However, since the environment variation is not complex when tracking on the plane, the force tracking errors for all five controllers are quite similar. Fig. 13 shows the force tracking performance for tracking the large deformation environment. Combining with Fig.

Table 2
MPRPO-DAIC parameters in simulation process.

Parameter name	Parameter symbol	Value
Mass coefficient	m_r	1 Ns^2/m
Damping coefficient	b_r	200 Ns/m
State weighted matrix	$\boldsymbol{W}_\xi$	diag(80, 80, 80000)
Cross weighted matrix	$\boldsymbol{W}_f$	diag(80, 80, 80000)
Input weighting coefficient	W_u	10
Optimization horizon	N_m	10
Prediction horizon	N_n	5

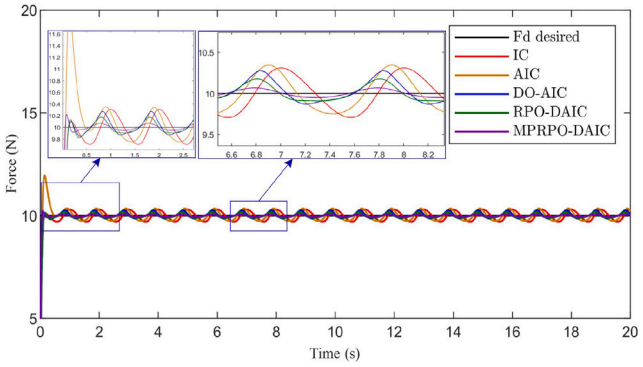


Fig. 12. Force tracking performance among different controllers over the plane.

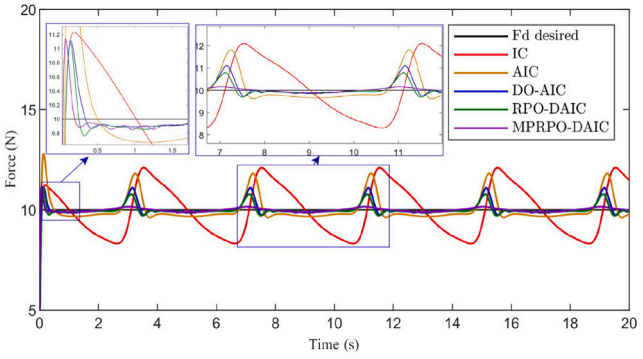


Fig. 13. Performance comparison of controllers in force tracking under a large deformation environment.

15, The force control accuracy of MPRPO-DAIC is observed to be 0.263 N, which improves by 62.70%, 57.65%, 37.97%, and 8.2% compared to IC, AIC, DO-AIC, and RPO-DAIC, respectively. Figs. 14 and 15 show that when tracking the high frequency changing environment, MPRPO-DAIC demonstrates the best force tracking performance, with a force control accuracy of 0.491 N, improving by 55.96%, 48.59%, 26.93%, and 16.35% compared to IC, AIC, DO-AIC, and RPO-DAIC, respectively. By comparing the controllers RPO-DAIC and DO-AIC, it is observed that rolling parameter optimization enhances the force tracking performance of DO-AIC to some extent. By comparing MPRPO-DAIC and RPO-DAIC, it is evident that the incorporation of model predictive control significantly improves the force tracking accuracy of RPO-DAIC in complex environments.

5. Complex surface tracking experiment

Fig. 16 shows a complex surface robotic force tracking platform. The robot used on the platform is a 7-DOF ROKAE xMate ER7 Pro, with a maximum payload of 7 kg and a repeatability of ±0.03 mm. The sampling rate of this robot is 1 ms. The workpieces tracked in the experiment are a large deformation surface and a high frequency changing surface. The sensor used is the KWR75B sensor, with a

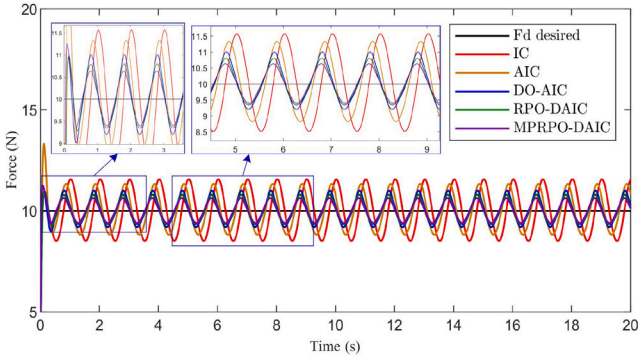


Fig. 14. Performance comparison of controllers in force tracking under a high frequency changing environment.

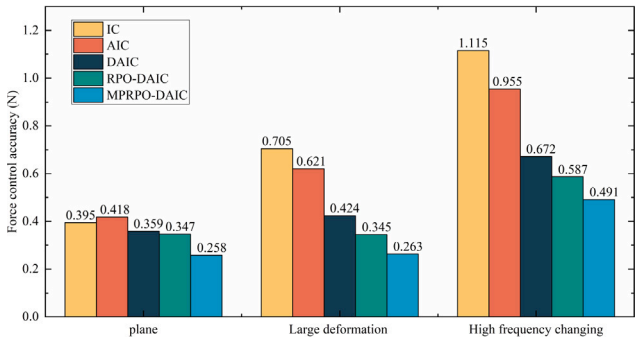


Fig. 15. Performance comparison of force control accuracy among controllers across three contact scenarios.

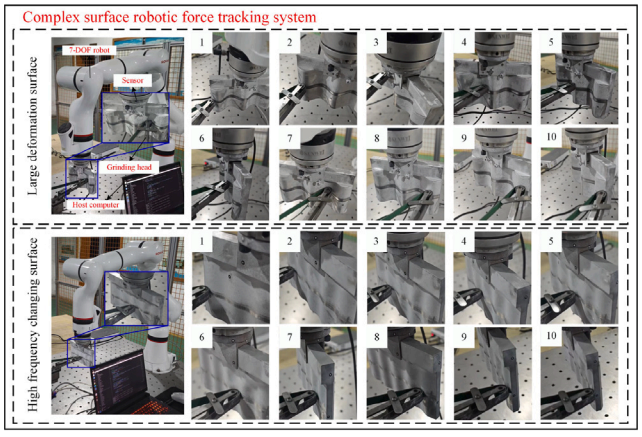


Fig. 16. Complex surface robotic force tracking system.

sampling frequency of 1000 Hz. Detailed information is provided in Table 3.

Force tracking experiments on complex surfaces were conducted by without controller, IC, AIC, DO-AIC, RPO-DAIC, and MPRPO-DAIC. AIC was proposed in Duan et al. (2018). The desired tracking force was set to $f_{Nd} = 5$ N. During the experiments, the mass coefficient m_r was set to 1. The damping coefficient b_r was configured to 400 for tracking the large deformation surface and 250 for tracking the high frequency changing surface. All other parameters were adjusted to achieve the best performance for each controller. The optimization horizon N_m in RPO-DAIC and MPRPO-DAIC was set to 10, and the prediction horizon N_n in MPRPO-DAIC was set to 5. These parameter settings were determined through empirical testing to balance control accuracy and real-time feasibility. It is worth noting that N_n should

Table 3
Detailed parameters of KWR75B sensor.

F_x, F_y (N)	F_z (N)	M_x, M_y (Nm)	M_z (Nm)	Sampling resolution (Bit)	Repetitive accuracy (%FS)	Precision (%FS)	Sampling frequency (Hz)
200	200	8	8	24	0.1	0.5	1000

not exceed 8, as the computation time of the prediction model exceeds 15 ms when the prediction 12 is set to 8, which leads to poor real-time performance.

The force tracking results of different controllers for the large deformation surface are shown in Fig. 17. It can be observed that MPRPO-DAIC effectively eliminates the force fluctuations caused by large surface deformations, demonstrating the best force control performance. Due to the presence of the dynamic observer, DO-AIC significantly reduces large force fluctuations compared to AIC. The rolling parameter optimization further enhances the performance of DO-AIC to some extent. Add MAE (Mean Absolute Error) as a supplementary performance metric to evaluate the force control error of the controller. As illustrated in Fig. 18(a), the force control accuracy of MPRPO-DAIC is 1.28 N, which represents improvements of 78.38%, 42.08%, 37.25%, 18.99%, and 11.72% compared to without controller, IC, AIC, DO-AIC, and RPO-DAIC, respectively. The force control mean error of MPRPO-DAIC is 1 N, which represents improvements of 81.41%, 41.18%, 35.48%, 15.97%, and 5.66% compared to without controller, IC, AIC, DO-AIC, and RPO-DAIC, respectively.

To evaluate the performance of each controller more comprehensively, two additional indicators, force control peak error and force control integrated error, are introduced. The former is obtained by calculating the maximum absolute deviation between the force signal and the target force, while the latter is based on the Integrated Squared Error (ISE) metric from Wang et al. (2021), which is normalized here to obtain the force control integrated error.

As illustrated in Fig. 18(b), the force control peak error of MPRPO-DAIC is 12.69 N, representing improvements of 46.26%, 45.78%, 16.15%, 33.33%, and 8.51% compared with the results of the system without controller, IC, AIC, DO-AIC, and RPO-DAIC, respectively. The peak error of DO-AIC is relatively large because it only reflects the maximum deviation at a single point and is more susceptible to occasional disturbances, although its overall trend is consistent with that of force control accuracy. The force control integrated error of MPRPO-DAIC is 35.05 N^2, representing improvements of 95.32%, 93.95%, 92.97%, 88.13%, and 85.99% compared with the results of the system without controller, IC, AIC, DO-AIC, and RPO-DAIC, respectively.

The force tracking results of different controllers for the high frequency changing surface are shown in Fig. 19. It can be observed that MPRPO-DAIC demonstrates the best force tracking performance during the process. The presence of the dynamic observer improves the robustness of AIC to some extent, while rolling parameter optimization further enhances the force control performance of DO-AIC. As shown in Fig. 20(a), the force control accuracy of MPRPO-DAIC is 0.81 N, representing improvements of 58.46%, 51.79%, 44.90%, 34.15%, and 31.36% compared to no controller, IC, AIC, DO-AIC, and RPO-DAIC, respectively. The force control mean error of MPRPO-DAIC is 0.65 N, which represents improvements of 58.86%, 52.55%, 44.92%, 33.67%, and 30.85% compared to without controller, IC, AIC, DO-AIC, and RPO-DAIC, respectively.

As illustrated in Fig. 20(b), the force control peak error of MPRPO-DAIC is 6.97 N, which represents improvements of 61.69%, 52.80%, 48.35%, 28.26%, and 13.20% compared to the systems without controller, and with IC, AIC, DO-AIC, and RPO-DAIC, respectively. The force control integrated error of MPRPO-DAIC is 3.82 N^2, representing improvements of 82.98%, 63.35%, 43.19%, 31.94%, and 25.65% compared to the same controllers. Considering all four force control evaluation metrics, MPRPO-DAIC demonstrates the best overall control performance in the robotic force tracking process on complex curved surfaces.

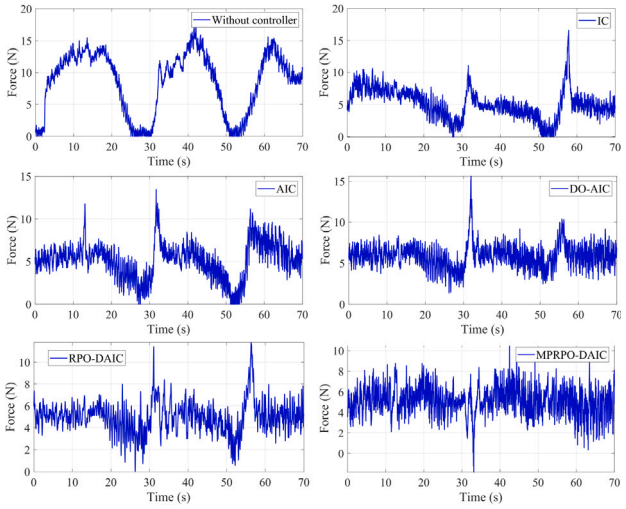
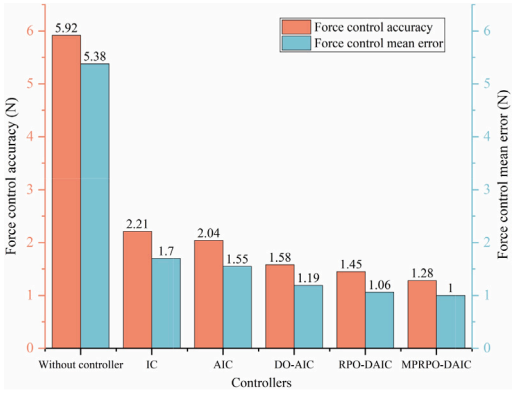
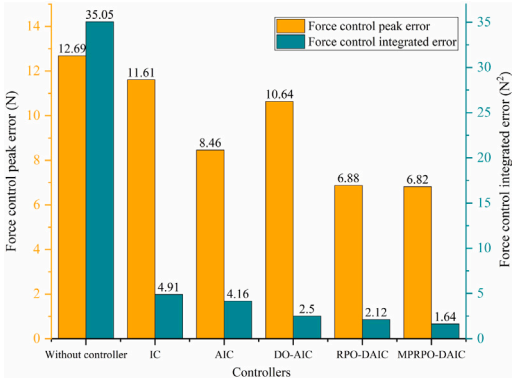


Fig. 17. Comparison of tracking force on the large deformation surface with various controllers.



(a) Comparison of force control accuracy and mean error (large deformation surface)



(b) Comparison of force control peak error and integrated error (large deformation surface)

Fig. 18. Comparison of force tracking performance on the large deformation surface with various controllers.

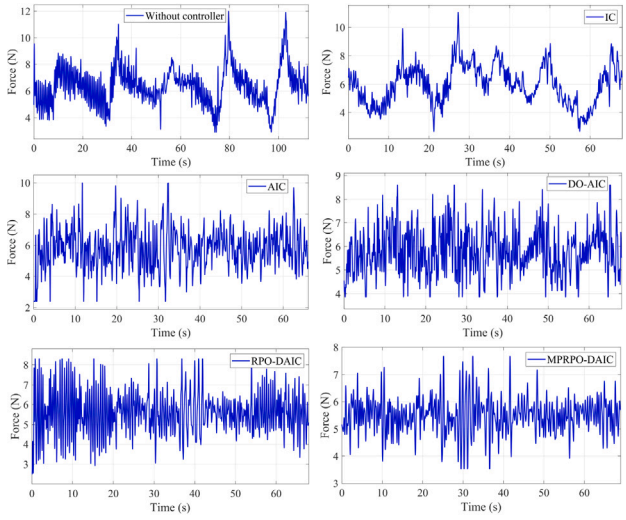
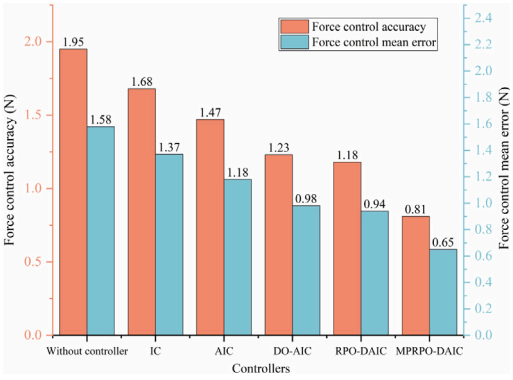
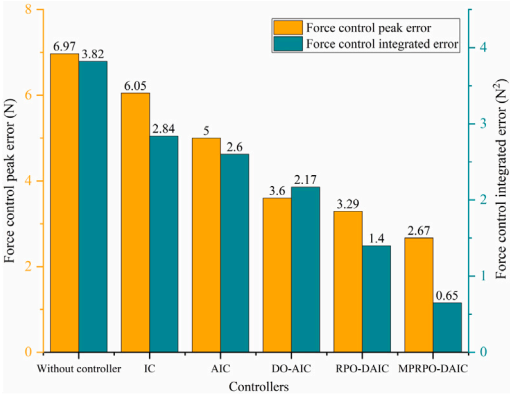


Fig. 19. Comparison of tracking force on the high frequency changing surface with various controllers.



(a) Comparison of force control accuracy and mean error (high-frequency changing surface)



(b) Comparison of force control peak error and integrated error (high-frequency changing surface)

Fig. 20. Comparison of force tracking performance on the high-frequency changing surface with various controllers.

From the above analysis, it can be seen that the presence of dynamic observer in DO-AIC effectively enhances the robustness of AIC when dealing with complex surfaces. Rolling parameter optimization of key parameters in the dynamic observer further improves the force control performance of DO-AIC. The force tracking performance of

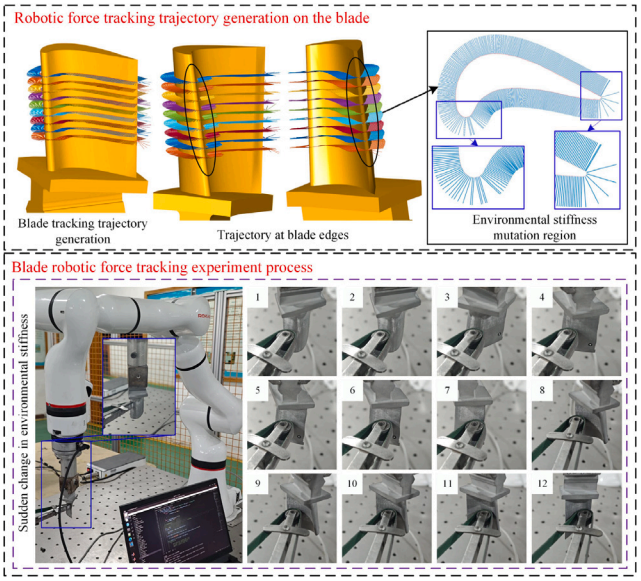


Fig. 21. Blade robotic force tracking experiment.

MPRPO-DAIC demonstrates that combining DO-AIC with model predictive control can significantly enhance robotic force tracking accuracy under complex surface conditions. The proposed MPRPO-DAIC algorithm has been validated for its effectiveness through tracking large deformation and high frequency changing surfaces.

To further verify the robustness of the MPRPO-DAIC in response to sudden changes in environmental stiffness, the robustness verification experiment was conducted on the blade, as shown in Fig. 21. Due to the significant curvature variation near the blade edge, abrupt changes in environmental stiffness often occur at the transitions between the concave, convex, and the leading and trailing edges during robotic grinding (Wang et al., 2024), resulting in drastic fluctuations in the tracking force. Force tracking experiments on turbine blades were carried out with AIC, RPO-DAIC, and MPRPO-DAIC, respectively. The controller parameters used in these tests were consistent with those employed in the complex surface force tracking experiments, and the desired tracking force was set to $f_{N_d} = 5$ N.

The force tracking results of different controllers for the blade are shown in Fig. 22. It can be observed that abrupt changes in environmental stiffness occur at the transition regions along the blade edge, leading to sudden variations in the force signal. Compared with AIC, both MPRPO-DAIC and RPO-DAIC effectively suppress these force fluctuations. Among them, MPRPO-DAIC exhibits better robustness than RPO-DAIC under such conditions. The detailed performance evaluation results of different controllers under sudden changes in environmental stiffness are presented in Table 4.

The results from robotic force tracking experiments on complex surfaces and blades suggest that MPRPO-DAIC exhibits promising force tracking performance in complex contact environments and is capable of maintaining good robustness under abrupt changes in environmental stiffness, providing valuable insights into the accurate control of grinding contact force in robotic grinding of complex curved components.

It is worth noting that this study primarily focuses on the field of robotic grinding and only investigates the regulation of normal contact force during the grinding process. Future work can extend the approach to multi-directional contact force control to enhance its applicability in more complex robotic contact scenarios.

Table 4
Force tracking performance on the blade with different controllers.

Controllers	Force control accuracy (N)	Force control mean error (N)	Force control peak error (N)	Force control integrated error (N^2)
AIC	2.93	2.03	14.73	8.58
RPO-DAIC	0.95	0.69	5.57	0.9
MPRPO-DAIC	0.83	0.57	5.28	0.69

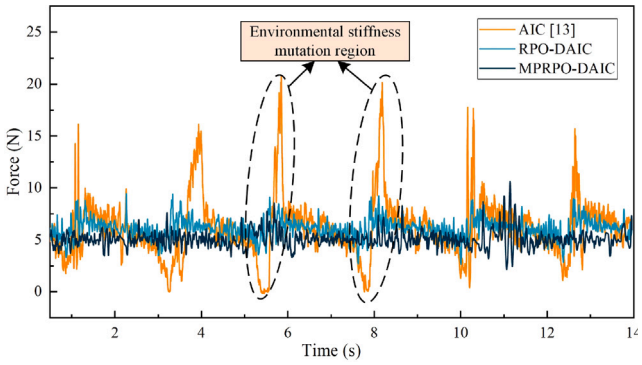


Fig. 22. Comparison of tracking force on the blade with various controllers.

6. Conclusion

This paper proposes a novel model predictive dynamic adaptive impedance control algorithm to achieve stable robotic contact force tracking on complex surfaces. A series of simulation and force tracking experiments are conducted to verify the effectiveness of the algorithm. The conclusions are summarized as follows:

(1) The dynamic adaptive impedance control algorithm is proposed, which incorporates a dynamic observer term into AIC to adjust the update rate in real-time based on historical information of contact force, thereby enhancing the robustness of AIC. Subsequently, the key parameters in the dynamic observer are optimized using the rolling parameter optimization algorithm, leading to an improvement in the contact force control performance of DO-AIC.

(2) The MPRPO-DAIC algorithm is developed by integrating the dynamic adaptive impedance control algorithm with model predictive control. A cost function is constructed with the objective of minimizing the contact force error, and the velocity compensation value is obtained by solving a quadratic programming problem. This velocity compensation value is used to adjust the robotic end-effector, enabling stable contact force control during complex surface tracking.

(3) Simulation and force tracking experiments on complex surfaces are conducted. In the simulation, the force control accuracy of MPRPO-DAIC improves by 57.65% and 48.59% over AIC for the large deformation and high frequency changing environments, respectively. In the force tracking experiments, the force control accuracy of MPRPO-DAIC improves by 37.25% and 44.90% compared to AIC for the large deformation and high frequency changing surfaces, respectively.

CRediT authorship contribution statement

Yilin Mu: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lai Zou:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Ziling Wang:** Methodology, Investigation, Funding acquisition. **Jiantao Li:** Methodology, Investigation, Formal analysis. **Linlin Jiang:** Validation. **Wenxi Wang:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Akdoğan, E., Aktan, M. E., Koru, A. T., Arslan, M. S., Atlıhan, M., & Kuran, B. (2018). Hybrid impedance control of a robot manipulator for wrist and forearm rehabilitation: Performance analysis and clinical results. *Mechatronics*, *49*, 77–91.

Arnold, J., Hanzlick, H., & Lee, H. (2019). Variable damping control of the robotic ankle joint to improve trade-off between performance and stability. In *2019 IEEE international conference on robotics and automation* (pp. 1699–1704).

Beck, F., Vu, M. N., Hartl-Nesic, C., & Kugi, A. (2024). Model predictive trajectory optimization with dynamically changing waypoints for serial manipulators. *IEEE Robotics and Automation Letters*, *9*, 6488–6495.

Bednarczyk, M., Omran, H., & Bayle, B. (2020). Model predictive impedance control. In *2020 IEEE international conference on robotics and automation* (pp. 4702–4708).

Chen, F., Zhao, H., Li, D., Chen, L., Tan, C., & Ding, H. (2018). Robotic grinding of a blisk with two degrees of freedom contact force control. *International Journal of Advanced Manufacturing Technology*, *101*, 461–474.

dos Santos, W. M., & Siqueira, A. A. G. (2019). Optimal impedance via model predictive control for robot-aided rehabilitation. *Control Engineering Practice*, *93*, Article 104177.

Duan, J., Gan, Y., Chen, M., & Dai, X. (2018). Adaptive variable impedance control for dynamic contact force tracking in uncertain environment. *Robotics and Autonomous Systems*, *102*, 54–65.

Gold, T., Völz, A., & Graichen, K. (2020). Model predictive interaction control for industrial robots. *IFAC-PapersOnLine*, *53*, 9891–9898.

Gold, T., Volz, A., & Graichen, K. (2023). Model predictive interaction control for robotic manipulation tasks. *IEEE Transactions on Robotics*, *39*, 76–89.

Hamedani, M. H., Sadeghian, H., Zekri, M., Sheikholeslam, F., & Keshmiri, M. (2021). Intelligent impedance control using wavelet neural network for dynamic contact force tracking in unknown varying environments. *Control Engineering Practice*, *113*, Article 104840.

Han, F., Tam, S., Cao, Z., Zhao, X., Tao, B., & Ding, H. (2023). Nonlinear impedance control with trajectory adaptation for collaborative robotic grinding. *Science China Technological Sciences*, *66*, 1928–1936.

Jin, L., Zhang, F., Liu, M., & Xu, S. S. D. (2023). Finite-time model predictive tracking control of position and orientation for redundant manipulators. *IEEE Transactions on Industrial Electronics*, *70*, 6017–6026.

Ke, X., Yu, Y., Li, K., Wang, T., Zhong, B., Wang, Z., et al. (2023). Review on robot-assisted polishing: Status and future trends. *Robotics and Computer-Integrated Manufacturing*, *80*, Article 102482.

Kim, S., Kim, J. P., & Ryu, J. (2014). Adaptive energy-bounding approach for robustly stable interaction control of impedance-controlled industrial robot with uncertain environments. *IEEE/ASME Transactions on Mechatronics*, *19*, 1195–1205.

Kleff, S., Dantec, E., Saurel, G., Mansard, N., & Righetti, L. (2022). Introducing force feedback in model predictive control. In *2022 IEEE/RSJ international conference on intelligent robots and systems* (pp. 13379–13385).

Labrecque, P., & Gosselin, C. (2018). Variable admittance for pHRI: From intuitive unilateral interaction to optimal bilateral force amplification. *Robotics and Computer-Integrated Manufacturing*, *52*, 1–8.

Li, D., Yang, J., Zhao, H., & Ding, H. (2022). Contact force plan and control of robotic grinding towards ensuring contour accuracy of curved surfaces. *International Journal of Mechanical Sciences*, *227*, Article 107449.

Liu, A., Yu, L., & Zhang, W. A. (2014). Moving horizon SINR estimation for wireless networked systems. *IEEE Transactions on Industrial Informatics*, *10*, 431–438.

Liu, A., Zhang, W. A., Chen, M. Z. Q., & Yu, L. (2017). Moving horizon estimation for mobile robots with multirate sampling. *IEEE Transactions on Industrial Electronics*, *64*, 1457–1467.

Minniti, M. V., Grandia, R., Fah, K., Farshidian, F., & Hutter, M. (2021). Model predictive robot-environment interaction control for mobile manipulation tasks. In *2021 IEEE international conference on robotics and automation* (pp. 1651–1657).

Mu, Y., Wang, Z., Zou, L., Wang, W., Lv, C., & Wang, A. (2023). A novel regional force control strategy based on seven-axis linkage grinding system to improve blade machining accuracy. *Journal of Manufacturing Processes*, *97*, 235–247.

Mu, Y., Zou, L., Wang, Z., Li, H., Yan, S., & Wang, W. (2025). A novel dynamic observer-based contact force control strategy in robotic grinding to improve blade profile accuracy. *Robotics and Computer-Integrated Manufacturing*, *94*, Article 102966.

Nurbayeva, A., & Rubagotti, M. (2025). Nonlinear model predictive control with set terminal constraint for safe robot motion planning via speed and separation monitoring. *Control Engineering Practice*, *154*, Article 106155.

Schiller, J. D., & Müller, M. A. (2023). Suboptimal nonlinear moving horizon estimation. *IEEE Transactions on Automatic Control*, *68*, 2199–2214.

Sheng, X., & Zhang, X. (2018). Fuzzy adaptive hybrid impedance control for mirror milling system. *Mechatronics*, *53*, 20–27.

Song, K., Xiao, G., Chen, S., Liu, X., & Huang, Y. (2023). A new force-depth model for robotic abrasive belt grinding and confirmation by grinding of the Inconel 718 alloy. *Robotics and Computer-Integrated Manufacturing*, *80*, Article 102483.

Wang, Z., Zou, L., Mu, Y., Li, H., Wang, W., & Huang, Y. (2023a). Region-based force control strategy for improving profile accuracy of blade on 7-axis linkage robotic grinding system. *IEEE/ASME Transactions on Mechatronics*, 1–12.

Wang, Z., Zou, L., Mu, Y., Li, H., Wang, W., & Huang, Y. (2023b). Region-based force control strategy for improving profile accuracy of blade on 7-axis linkage robotic grinding system. *IEEE/ASME Transactions on Mechatronics*, 1–12.

Wang, Z., Zou, L., Su, X., Luo, G., Li, R., & Huang, Y. (2021). Hybrid force/position control in workspace of robotic manipulator in uncertain environments based on adaptive fuzzy control. *Robotics and Autonomous Systems*, *145*, Article 103870.

Wang, Z., Zou, L., Zhang, J., Li, H., Wang, W., & Huang, Y. (2024). Tool axis vector optimization for robotic grinding based on measured point cloud of complex curved blade. *Advanced Engineering Informatics*, *62*, Article 102716.

Wijayarathne, L., Zhou, Z., Zhao, Y., & Hammond, F. L. (2023). Real-time deformable-contact-aware model predictive control for force-modulated manipulation. *IEEE Transactions on Robotics*, *39*, 3549–3566.

Wu, J., Ni, F., Zhang, Y., Fan, S., Zhang, Q., Lu, J., et al. (2018). Smooth transition adaptive hybrid impedance control for connector assembly. *Industrial Robot: An International Journal*, *45*, 287–299.

Xu, X., Chen, W., Zhu, D., Yan, S., & Ding, H. (2021a). Hybrid active/passive force control strategy for grinding marks suppression and profile accuracy enhancement in robotic belt grinding of turbine blade. *Robotics and Computer-Integrated Manufacturing*, *67*, Article 102047.

Xu, X., Chen, W., Zhu, D., Yan, S., & Ding, H. (2021b). Hybrid active/passive force control strategy for grinding marks suppression and profile accuracy enhancement in robotic belt grinding of turbine blade. *Robotics and Computer-Integrated Manufacturing*, *67*, Article 102047.

Xu, X., Zhu, D., Zhang, H., Yan, S., & Ding, H. (2019a). Application of novel force control strategies to enhance robotic abrasive belt grinding quality of aero-engine blades. *Chinese Journal of Aeronautics*, *32*, 2368–2382.

Xu, X., Zhu, D., Zhang, H., Yan, S., & Ding, H. (2019b). Application of novel force control strategies to enhance robotic abrasive belt grinding quality of aero-engine blades. *Chinese Journal of Aeronautics*, *32*, 2368–2382.

Yun, J., Sun, Y., Li, C., Jiang, D., Tao, B., Li, G., et al. (2022). Self-adjusting force/bit blending control based on quantitative factor-scale factor fuzzy-PID bit control. *Alexandria Engineering Journal*, *61*, 4389–4397.

Zhang, H., Li, L., Zhao, J., & Zhao, J. (2021). The hybrid force/position anti-disturbance control strategy for robot abrasive belt grinding of aviation blade base on fuzzy PID control. *International Journal of Advanced Manufacturing Technology*, *114*, 3645–3656.

Zhang, H., Li, L., Zhao, J., Zhao, J., Liu, S., & Wu, J. (2020). Design and implementation of hybrid force/position control for robot automation grinding aviation blade based on fuzzy PID. *International Journal of Advanced Manufacturing Technology*, *107*, 1741–1754.

Zhang, Y., Zhao, X., Chen, Y., Tao, B., & Ding, H. (2024). Observer-based variable impedance control using moving horizon estimation for robot machining thin-walled workpieces. *IEEE Transactions on Industrial Electronics*, *71*, 5972–5982.

Zhou, B., & Tian, T. (2024). Robotic disc grinding path planning method based on multi-objective optimization for nuclear reactor coolant pump casing. *Journal of Manufacturing Systems*, *77*, 810–833.

Zhu, G., Cai, M., Wang, Z., Li, L., Zhao, J., & Zhou, B. (2023). A new calibration method for a dynamic coordinate system in a robotic blade grinding and polishing system based on the six-point limit principle. *Robotics and Computer-Integrated Manufacturing*, *83*, Article 102561.

Zhu, H., Liu, M., & Yu, D. (2023). Learning-based model predictive control with application in robotic trajectory tracking. In *2023 62nd IEEE conference on decision and control* (pp. 2596–2601).